313326

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3 Hours / 70 Marks Seat No. | | | | | | | | |

Instructions :

(1) All Questions are *compulsory.*

(2) Illustrate your answers with neat sketches wherever necessary.

(3) Figures to the right indicate full marks.

(4) Assume suitable data, if necessary.

(5) Use of Non-programmable Electronic Pocket Calculator is permissible.

(6) Mobile Phone, Pager and any other Electronic Communication devices are not permissible in Examination Hall.

Marks

1. Attempt any FIVE of the following : 10

(a) Define simplex and quadra duplex system with neat sketch.

(b) Define the term signal to noise ratio.

(c) State formula to calculate bandwidth of AM signal.

(d) Draw FM wave in time domain and frequency domain.

(e) State the need of modulation in communication system (any four).

(f) Write the value of IF in :

(i) FM radio receiver

(ii) MW band in AM

(g) Draw sketch of Loop Antenna along with its radiation pattern.

2. Attempt any THREE of the following : 12

(a) Compare characteristic of asynchronous and synchronous transmission mode (any four pt.)

(b) Justify electromagnetic wave is said to be transverse wave.

(c) Compare AM and FM w.r.t. following points :

(i) Definition

(ii) Modulation Index

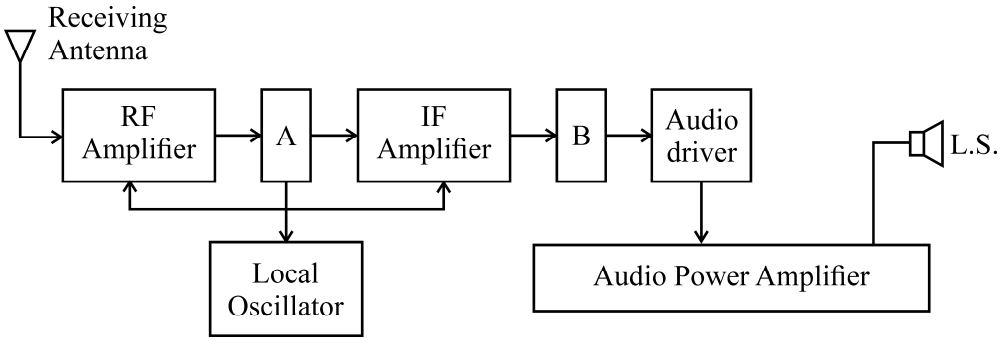
(iii) Bandwidth

(iv) Application

(d) Draw block diagram of PLL as FM demodulator and describe function of each block.

3. Attempt any THREE of the following : 12

(a) Redraw the block by identifying the blank blocks. Explain the role of A and B block. (Fig. – 1)



(Fig. – 1)

(b) Differentiate between ground wave and sky wave propagation (any four points).

(c) Find out type of propagation of following applications :

(i) AM radio broadcasting

(ii) Ship to shore propagation

(iii) Microwave links

(iv) Satellite communication

(d) Draw neat sketch of Yagi-Uda antenna and its radiation pattern. State function of each element in antenna.

4. Attempt any THREE of the following : 12

(a) Draw and explain block diagram of high level AM transmitter.

(b) Differentiate between FM and PM to the following parameters :

(i) Definition

(ii) Noise

(iii) Constant Parameter

(iv) Waveforms

(c) Explain sky wave propagation with neat sketch.

(d) Define following terms :

(i) Virtual height

(ii) Actual height

(iii) Critical frequency

(iv) Maximum usable frequency

(e) Compare resonant and non-resonant antenna on the basis of :

(i) Definition

(ii) Circuit

(iii) Reflection coefficient

(iv) Radiation pattern

5. Attempt any TWO of the following : 12

(a) Write down range of different frequencies in electromagnetic spectrum and one application area of each for following frequency :

(i) Voice frequency

(ii) High frequency

(iii) Infrared frequency

(iv) Visible spectrum

(v) Radio frequency

(vi) UV frequency

(b) Define modulation index of AM wave. Explain effect of modulation index on AM wave with waveforms.

(c) Explain the concept of pre-emphasis and de-emphasis with circuit diagram.

6. Attempt any TWO of the following : 12

(a) If there are three AM transmitter named A, B and C using following specifications :

- A transmitter having P_C = 1000 W and modulation index m = 50%.
- B transmitter having P_C = 1200 W and modulation index m = 60%
- C transmitter having P_C = 800 W and modulation index m = 80%

Find :

(i) Transmitted power of A, B, C.

(ii) Justify whether A, B & C is better as far as transmitted power.

(b) Draw constructional details of dish antenna. Describe its operating principle. Also draw its radiation pattern.

(c) Write two applications of following antenna :

(i) Yagi-Uda antenna

(ii) Microwave antenna

(iii) Dish antenna

(iv) Horn antenna

(v) Micro strip patch antenna

(vi) Smart antenna

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